

Figure 3: Block diagram of LVM2, EVMS and dependency on Device Mapper in Linux

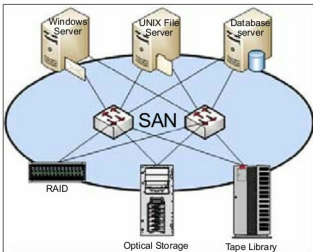


Figure 4: Network-based virtualisation

data ages, it can be stored on archival systems that could take a longer time to retrieve (See Figure 5). By doing this, data centre designers can optimise the price-performance-capacity equation. This hierarchical storage management was in operation even before virtualisation came into force. By virtualising the storage, it could be simpler to completely automate this process.

In an enterprise, the applications are tiered based on business criticality. Applications such as Enterprise Resource Planning and Customer Order Management are the most important and highly demanding applications. They need storage types such as SSD to meet their application needs.

Applications that assist in business decisions such as analytics could be the second tier that would need storage with lower latency, but the media type to be used would be based on a cost-benefit analysis. Enterprise applications such as Content Management Systems could fall into a category where the capacity is of a higher priority compared to I/O performance. Such data is typically stored on media such as discs with RAID capabilities. Email is an application type that could fall into both business critical services as well as capacity intensive applications.

For statutory reasons and also to maintain the history



Figure 5: The applications pyramid based on the demand for storage performance or capacity

of business, some data needs to be archived. The need for retrieving such data is rare and a slight delay in its retrieval does not adversely impact business. Such data is typically archived either on tapes or virtual tapes.

Let us now understand some key technologies that drive data centres to full virtualisation.

## Data deduplication

Deduplication is the process of eliminating redundancy of data, thus reducing the number of storage devices. This not only has an impact on capacity, but also network bandwidth in case of data being backed up over the network. There are two types of deduplication – inline and post-process. Inline deduplication kicks in before data is written on to the disk. Post-process deduplication analyses the data at a later time without interfering during the write process. Each has its own advantages and disadvantages.

## Thin provisioning

Traditional provisioning requires the full disk space to be available when configuring capacity for an application. There is no definite way to overcome the dilemma of how much of extra space needs to be allocated for future growth. Hence, administrators typically allocate storage space using a wild guess. This full allocation not only leaves unused capacity unavailable to other applications, but also increases the operational costs of keeping the storage devices running.

In contrast, thin provisioning allows administrators to allocate disk space that is even greater than the current capacity available in the data centre. This also works great when used in conjunction with the storage configurations of multiple applications. It is just like the over-subscription of network bandwidth that telecom operators practice or what the insurance companies do. The actual allocation of disc space happens just-in-time and hence there is no hogging of unused disk space by a few applications.

Thin provisioning does come with a few limitations. One of the scenarios for which thin provisioning is not recommended is when applications expect the data to be in contiguous blocks for I/O optimisation.